Introduction to Linux Systems

Version Control System: Git

Lab2: Create your Git account and first project

Chia-Heng Tu
Dept. of Computer Science and Information Engineering
National Cheng Kung University Fall 2025

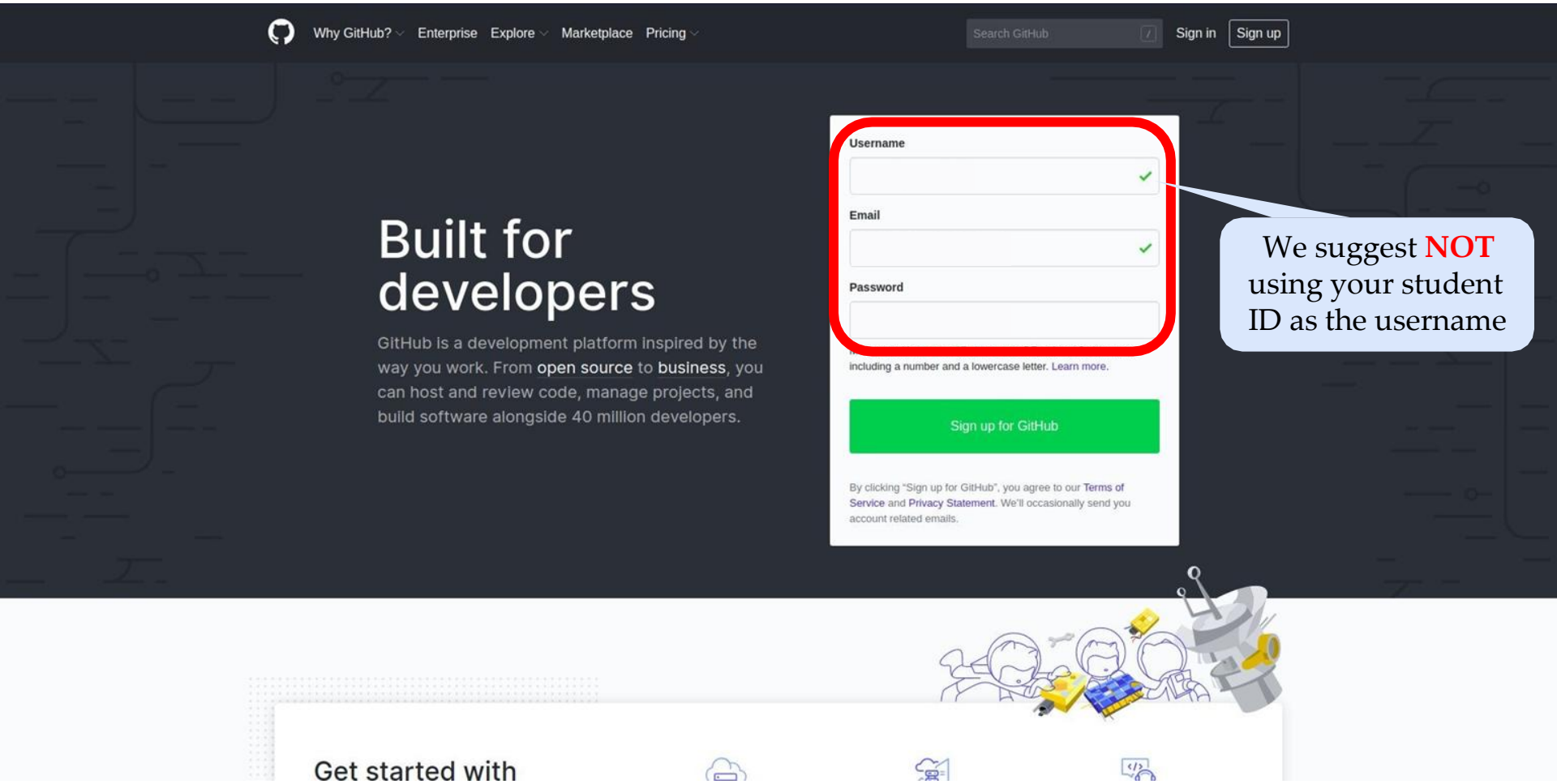
Lab

- Register the GitHub account
- Fork a repository and clone it to the local
- Practice the GitHub flow

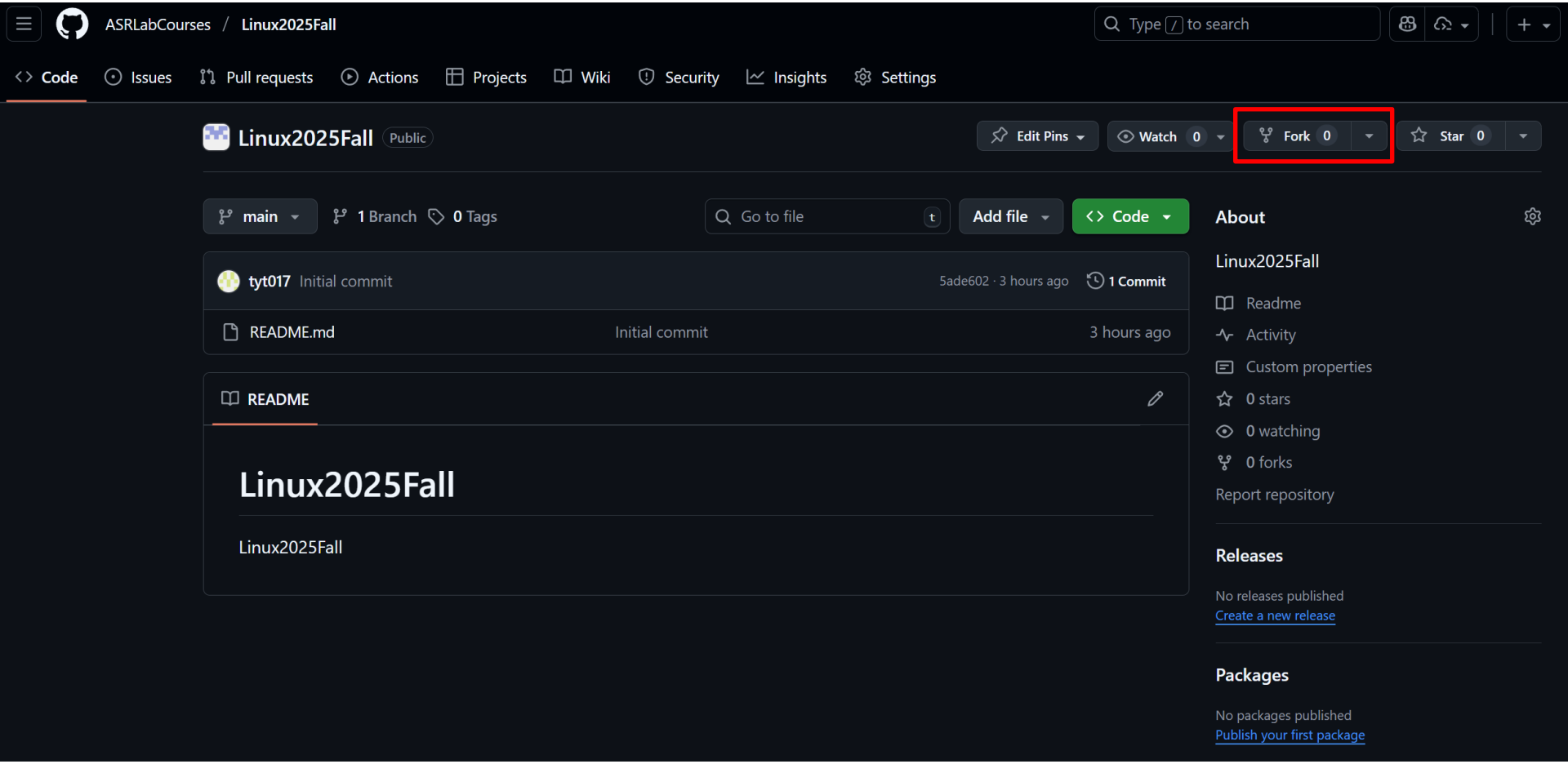
Install git on Linux

```
$ sudo apt install git
```

Register an account on GitHub (https://github.com/)

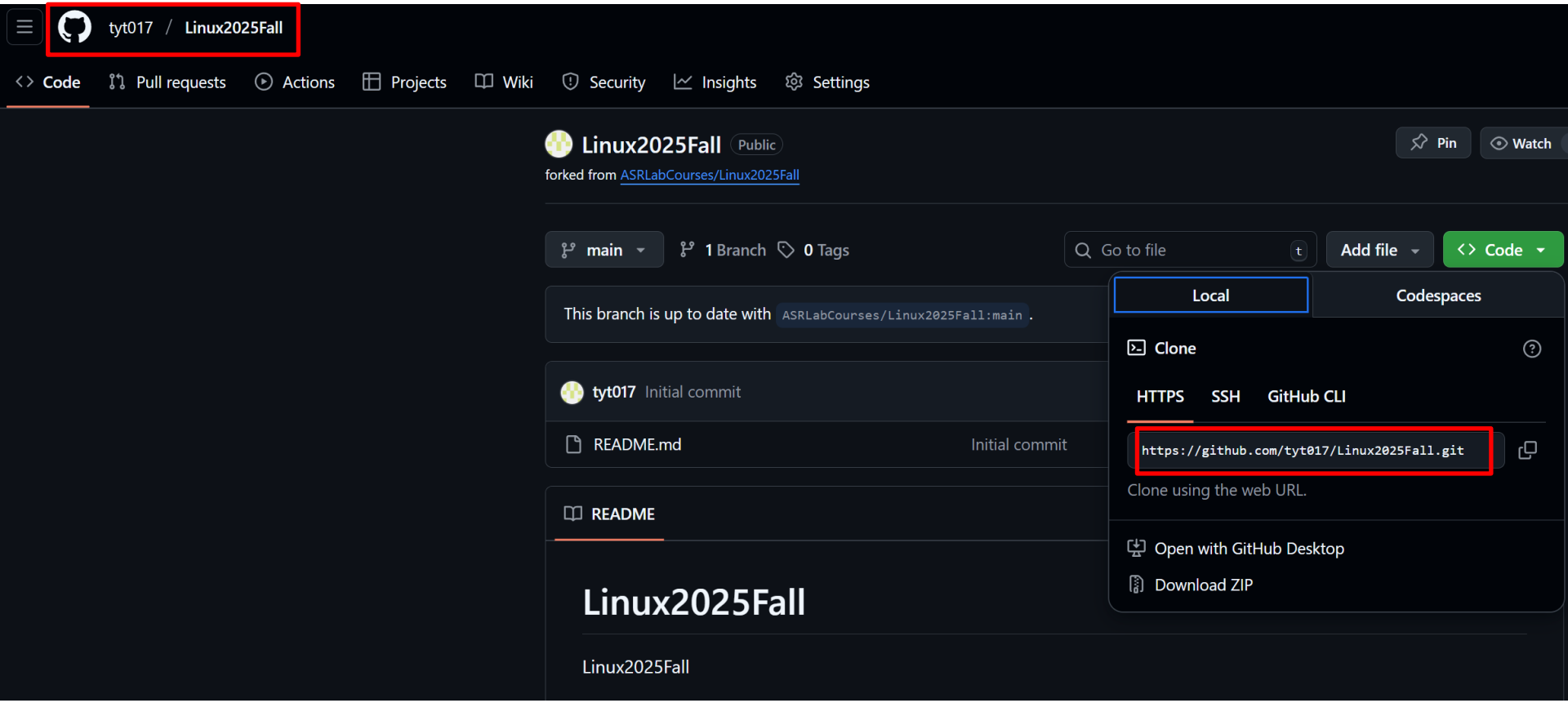


Fork a repository from "https://github.com/ASRLabCourses/Linux2025Fall"



第一種clone方式: 透過 http
Copy the URL of forked repo

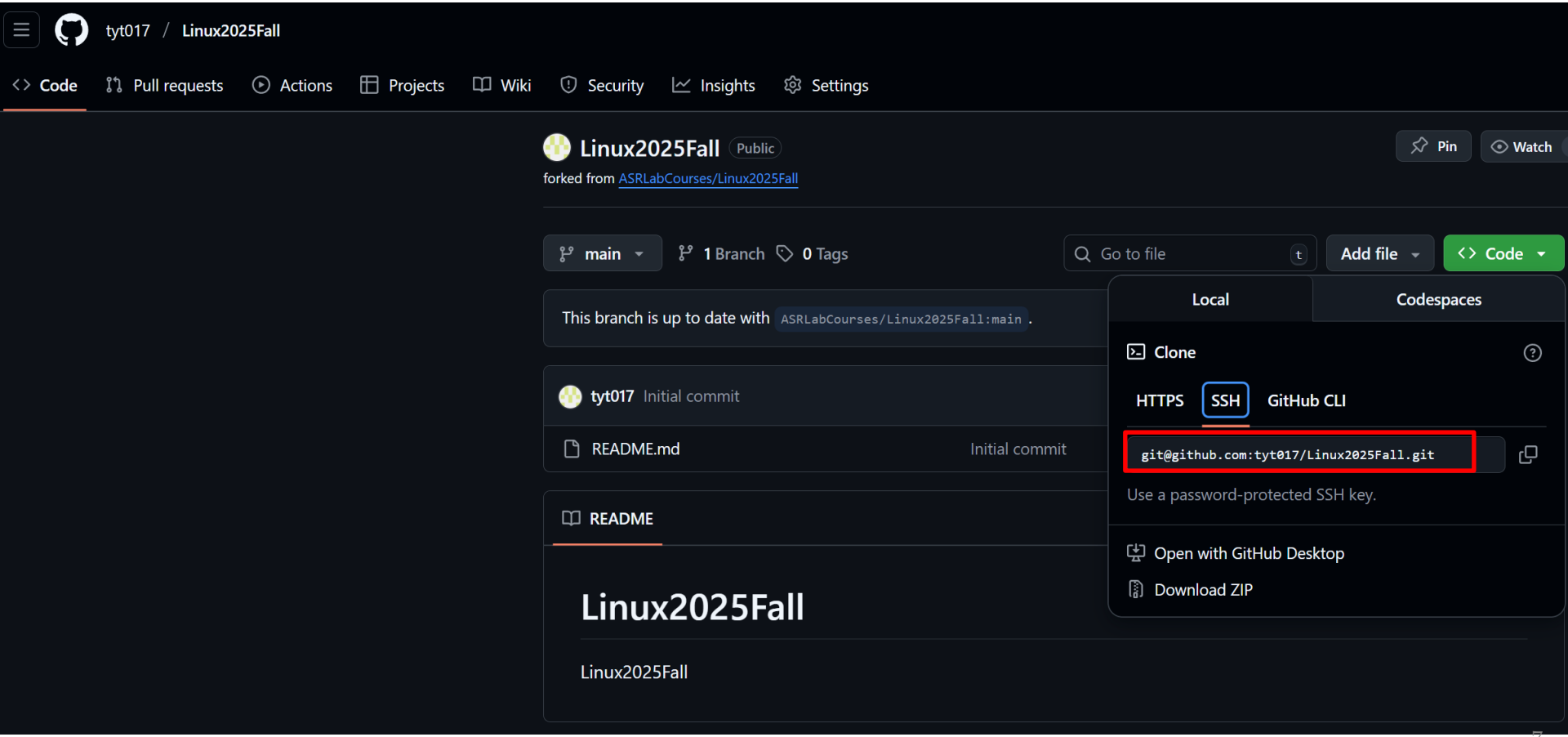
fork 後的 repo 會是 username / Linux2025Fall (點擊左邊主選單，於左列找到 <Username>/Linux2025Fall)



注意！是要複製剛剛 fork 的 repo.，而非 https://github.com/ASRLabCourses/Linux2025Fall.git

第二種clone的方式: 透過ssh
SSH Key - clone repo.

如果選擇使用 ssh key，clone repo. 要選擇 ssh 而非http



Authentication

以下兩種方式擇一去push 即可

1. Personal Access Token (recommended for http method)
2. SSH Key (recommended for ssh method)

- Generate ssh key
- Add ssh key to Github

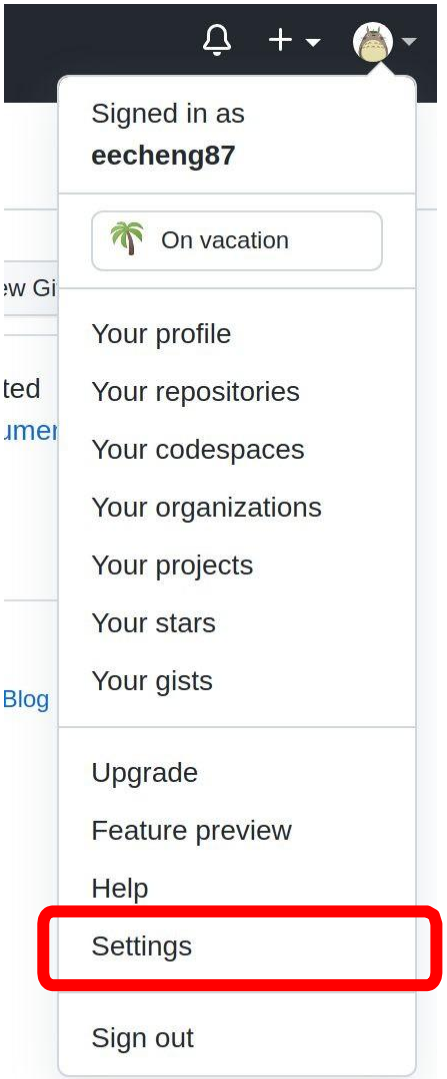
1. Personal access token

Settings -> Developer Settings

-> Personal Access Token

-> Tokens (classic)

-> Generate new token (classic)



1. Personal access token (cont.)

New personal access token

Personal access tokens function like ordinary OAuth access tokens. They can be used instead of a password for Git over HTTPS, or can be used to authenticate to the API over Basic Authentication.

Note

What's this token for?

Expiration *

30 days | The token will expire on Mon, Oct 25 2021

Select scopes

Scopes define the access for personal tokens. Read more about OAuth scopes.

repo	Full control of private repositories
repo:status	Access commit status
repo_deployment	Access deployment status
public_repo	Access public repositories
repo:invite	Access repository invitations
security_events	Read and write security events
workflow	Update GitHub Action workflows
write:packages	Upload packages to GitHub Package Registry
read:packages	Download packages from GitHub Package Registry

1. Personal access token (cont.)

☐ read:user	Read ALL user profile data
☐ user:email	Access user email addresses (read-only)
☐ user:follow	Follow and unfollow users
☐ **delete_repo**	Delete repositories
☐ **write:discussion**	Read and write team discussions
☐ read:discussion	Read team discussions
☐ **admin:enterprise**	Full control of enterprises
☐ manage_billing:enterprise	Read and write enterprise billing data
☐ read:enterprise	Read enterprise profile data
☐ **admin:gpg_key**	Full control of public user GPG keys (Developer Preview)
☐ write:gpg_key	Write public user GPG keys
☐ read:gpg_key	Read public user GPG keys

Generate token　Cancel

接著畫面會跳轉，會出現類似一串

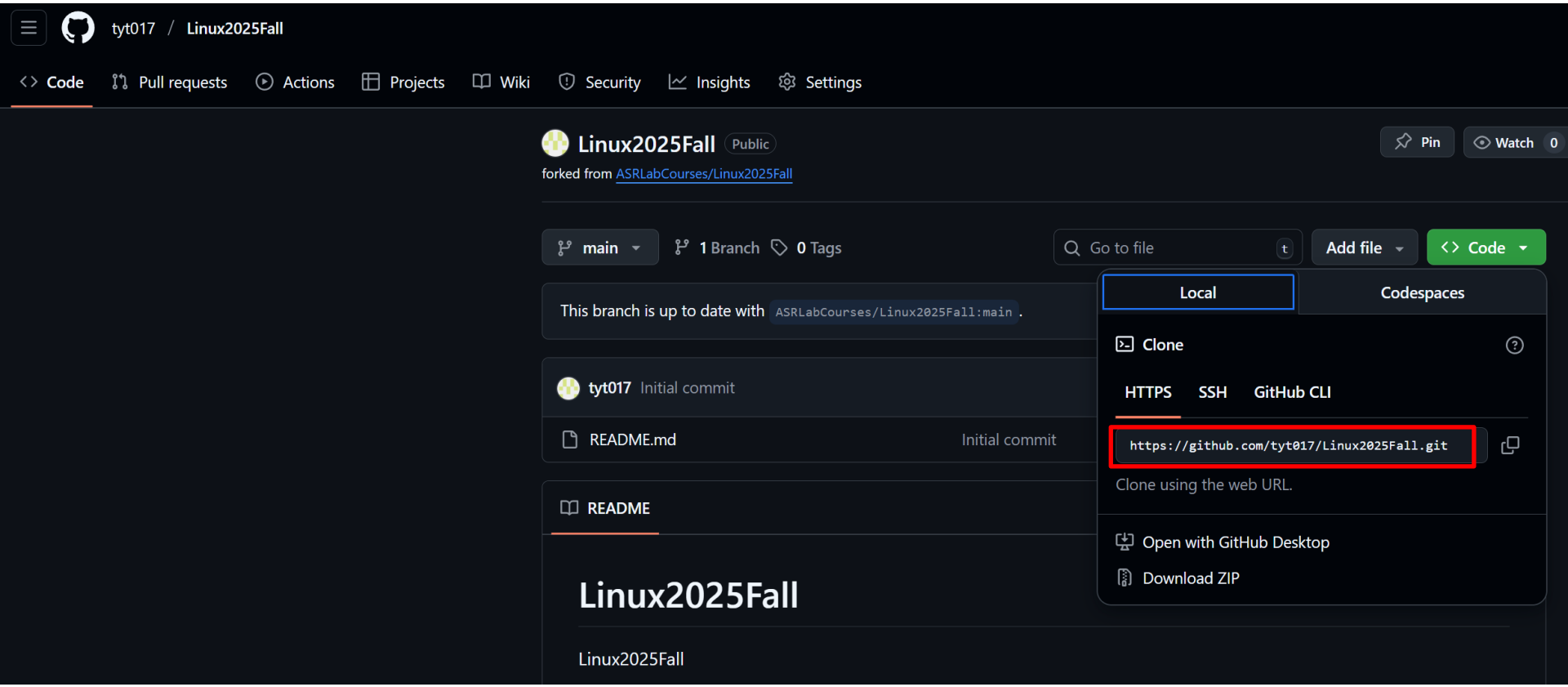
`ghp_sFhFsSHhTzMDreGRLjmks4Tzuzgthdvfsrta`

的 token，建議存起來，因為它只會出現一次

1. Clone forked repo to the local

$ git clone https://github.com/<Your username>/Linux2025Fall.git

$ cd Linux2025Fall/



2. SSH Key - Generate ssh key

$ ssh-keygen

一直按 enter 用預設即可

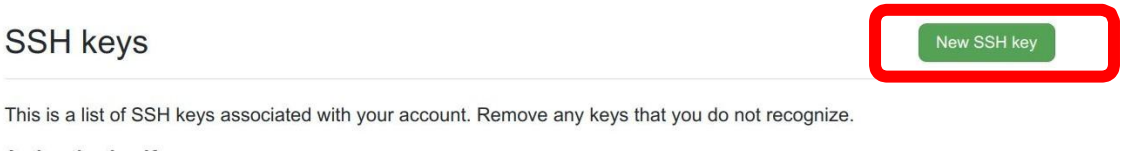
$ cat ~/.ssh/id_rsa.pub

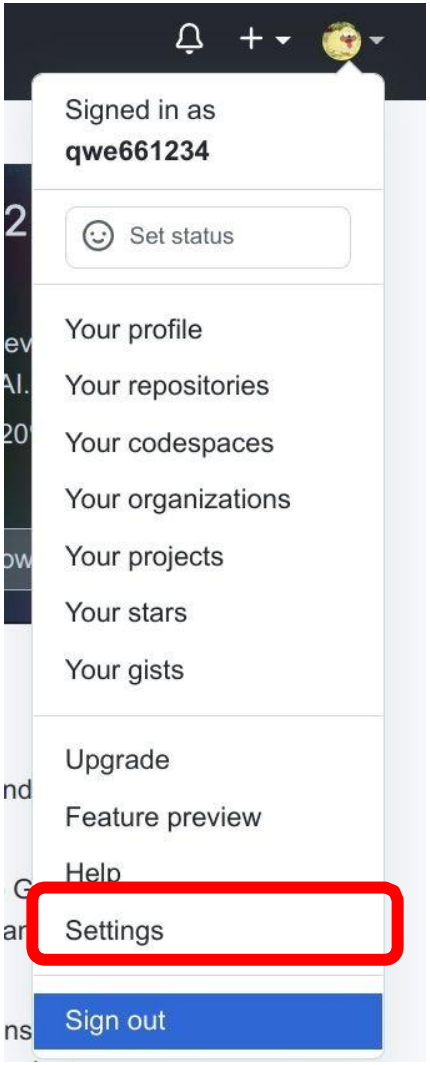
預期會得到 ssh-rsa XXXXXXX (一串token)，接著請複製這串 token　　(ssh-rsa XXXXXXX)

2. SSH Key - Add ssh key to Github

Settings -> SSH and GPG keys

-> New SSH key





2. SSH Key - Add ssh key to Github

SSH keys / Add new

Title

自定義 ssh key 的 title

Key type

Authentication Key

Key

貼上剛剛複製的那串 ssh token

Begins with 'ssh-rsa', 'ecdsa-sha2-nistp256', 'ecdsa-sha2-nistp384', 'ecdsa-sha2-nistp521', 'ssh-ed25519', 'sk-ecdsa-sha2-nistp256@openssh.com', or 'sk-ssh-ed25519@openssh.com'

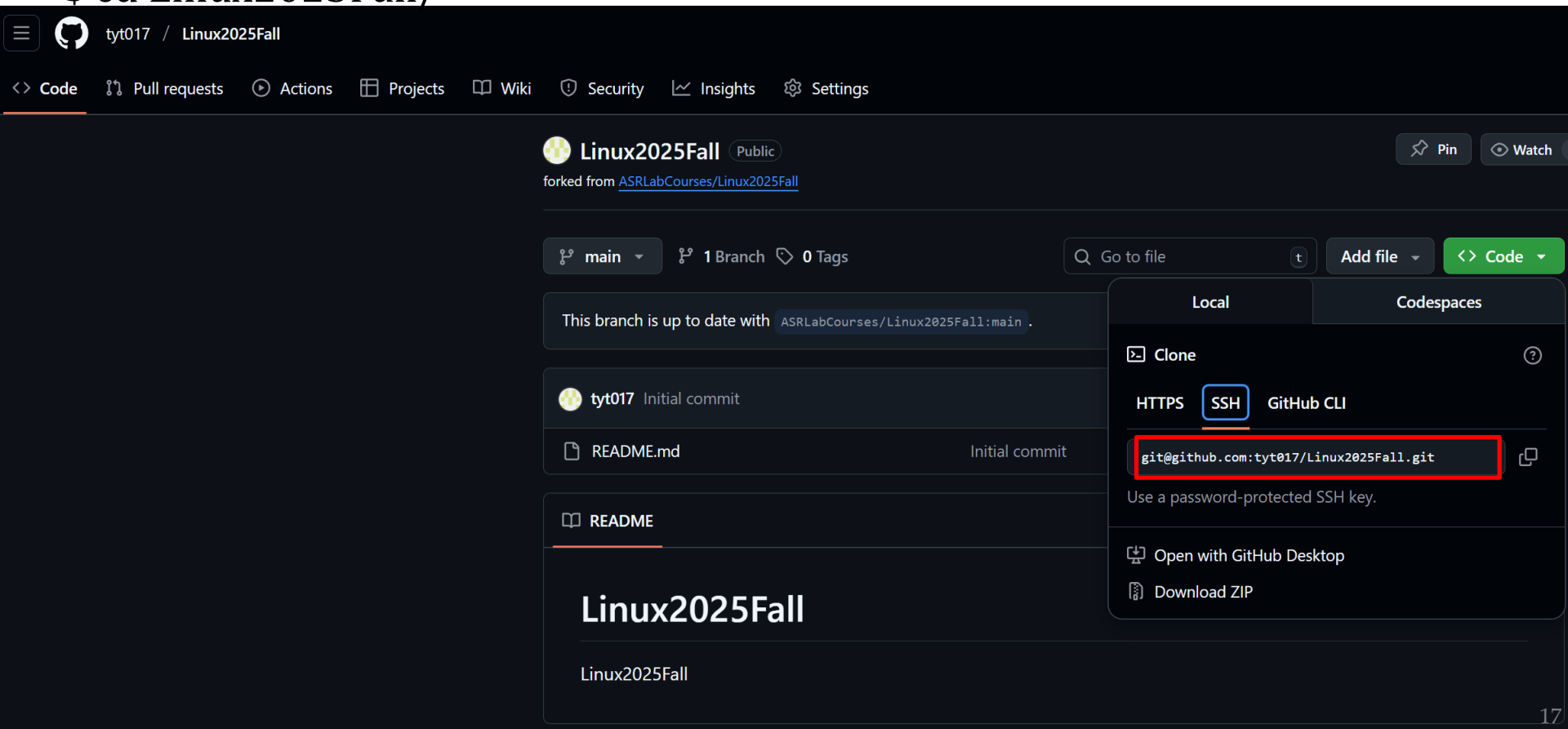
Add SSH key

新增成功後 SSH keys 會新增剛剛輸入的 Title 的 Authentication Keys

2. Clone forked repo to the local

$ git clone git@github.com:<Your username>/Linux2025Fall.git

$ cd Linux2025Fall/



Configure personal information

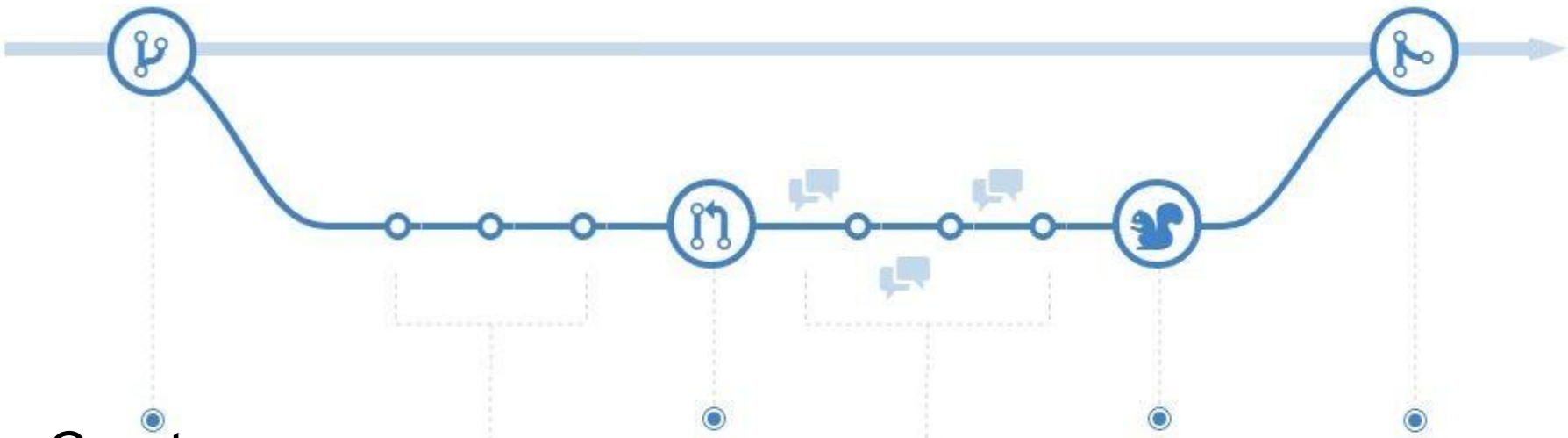
```
$ git config --global user.name "your_github_username"
$ git config --global user.email "your_github_email"
$ git config -l
```

“git config –l” command 預期輸出

user.email = <your_github_email>

user.name = <your_github_username>

GitHub Flow



Create a branch

Add commits

Open a Pull Request

Discuss and code review

Deploy or testing

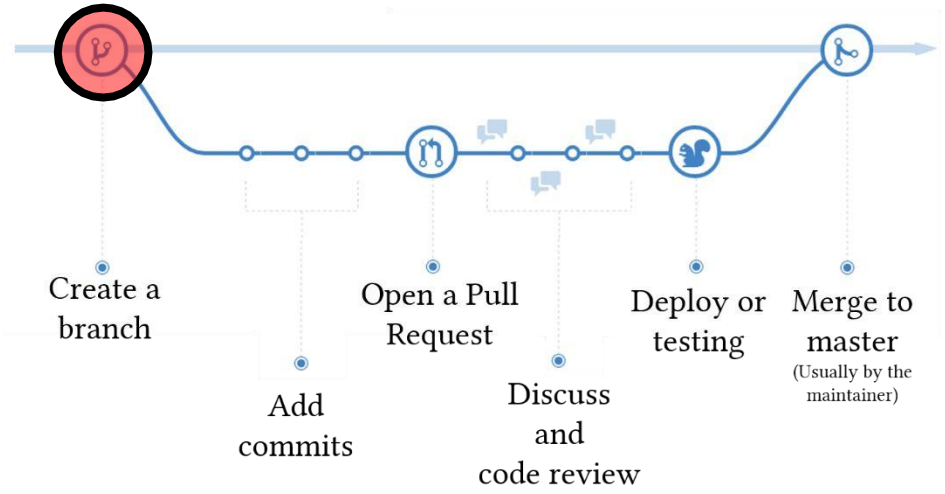
Merge to master
(Usually by the maintainer)

https://guides.github.com/introduction/flow/

Create a new branch

$ git checkout -b lab-2-git

可用 $ git branch 檢查有沒有成功創建新的 branch



Create a new file with content

```
$ mkdir -p f12345678/lab_2_git
$ cd f12345678/lab_2_git
$ echo "first text" > first.txt
```

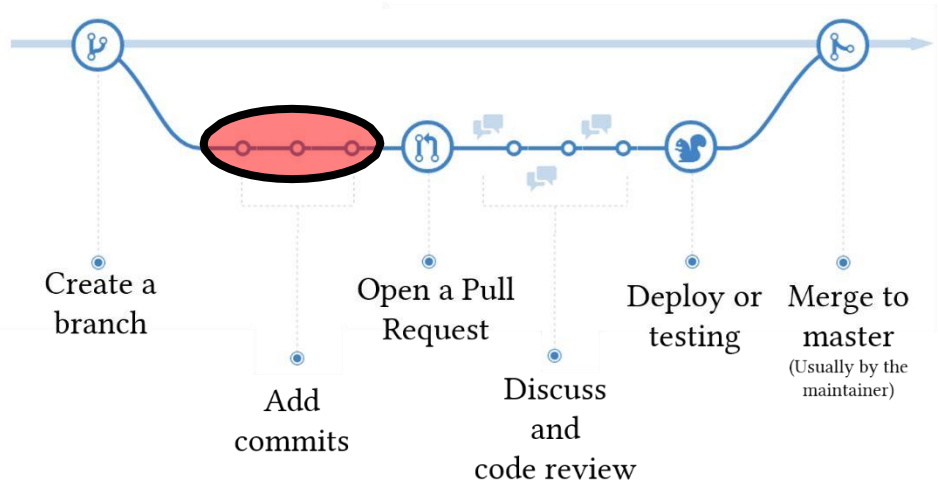
Change f12345678 to your student ID

Track the modification

```
$ git add first.txt
$ git status
$ git commit -m "Add F12345678"
```

Check new file tracked

Set the **name** and **email** of the user first by git config before commit command.



1. Push the branch to GitHub (Personal Access Token)

$ git push origin lab-2-git

remote

branch

分別輸入你的帳號和剛剛得到的 token

```
tyt@tyt-PC:~/Desktop/Linux2024Fall/[redacted]/lab_2_git$ git push origin lab-2-git
Username for 'https://github.com': tyt017
Password for 'https://tyt017@github.com':
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 12 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (5/5), 382 bytes | 382.00 KiB/s, done.
Total 5 (delta 0), reused 0 (delta 0)
remote:
remote: Create a pull request for 'lab-2-git' on GitHub by visiting:
remote:      https://github.com/tyt017/Linux2024Fall/pull/new/lab-2-git
remote:
To https://github.com/tyt017/Linux2024Fall.git
 * [new branch]      lab-2-git -> lab-2-git
```

2. Push the branch to GitHub (SSH Key)

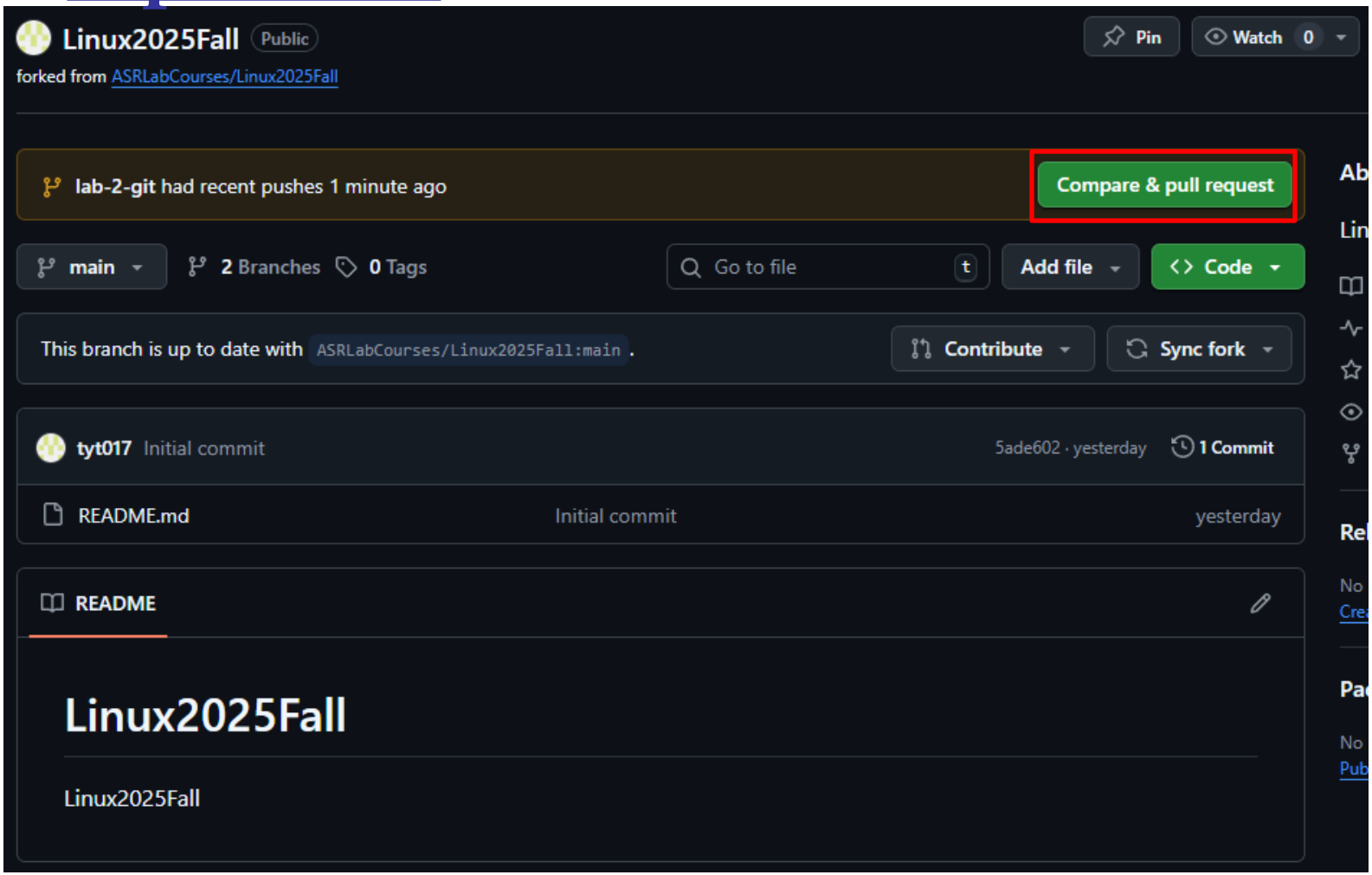
$ git push origin lab-2-git

remote branch

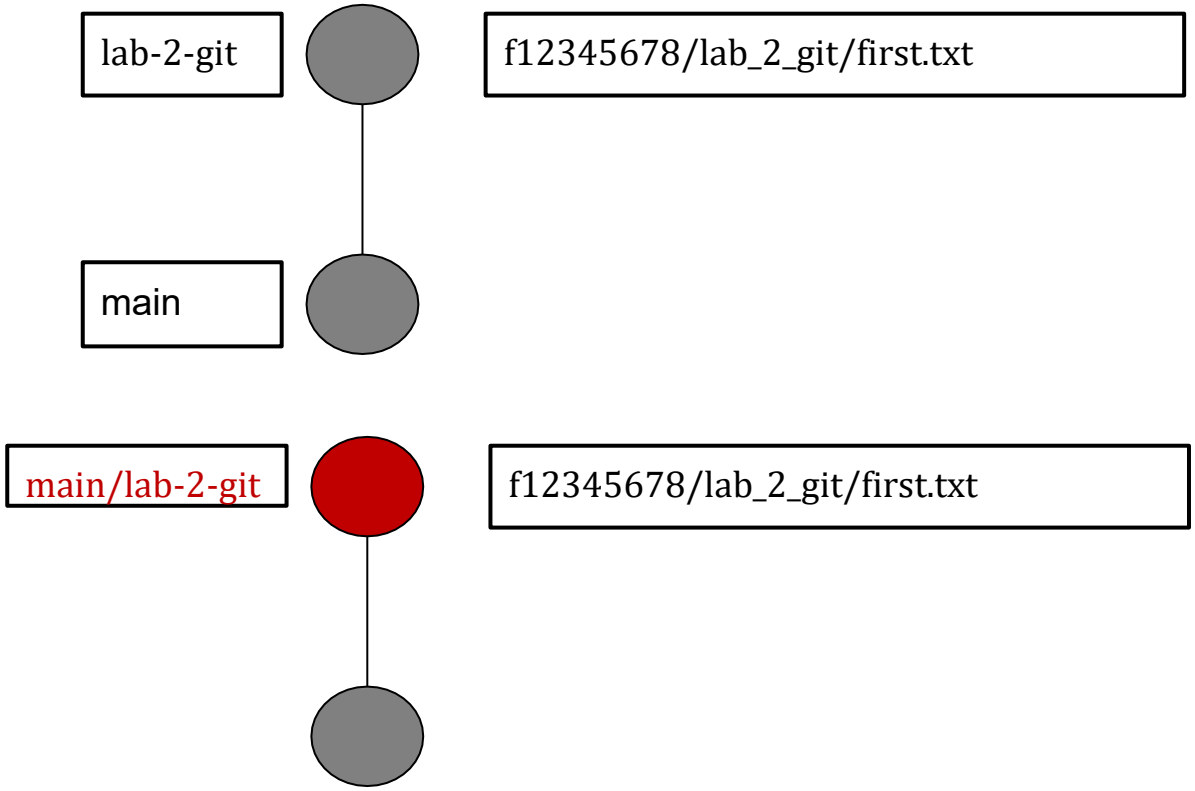
```
qwe661234@qwe661234-linux-2023:~/Linux2023Fall/f12345678$ git push origin lab-2-git

Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 64 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (4/4), 333 bytes | 333.00 KiB/s, done.
Total 4 (delta 0), reused 0 (delta 0)
remote:
remote: Create a pull request for 'lab-2-git' on GitHub by visiting:
remote:      https://github.com/qwe661234/Linux2023Fall/pull/new/lab-2-git
remote:
To github.com:qwe661234/Linux2023Fall.git
 * [new branch]      lab-2-git -> lab-2-git
```

Create pull-request on the web UI (optional)



Merge your branch in local and push to remote



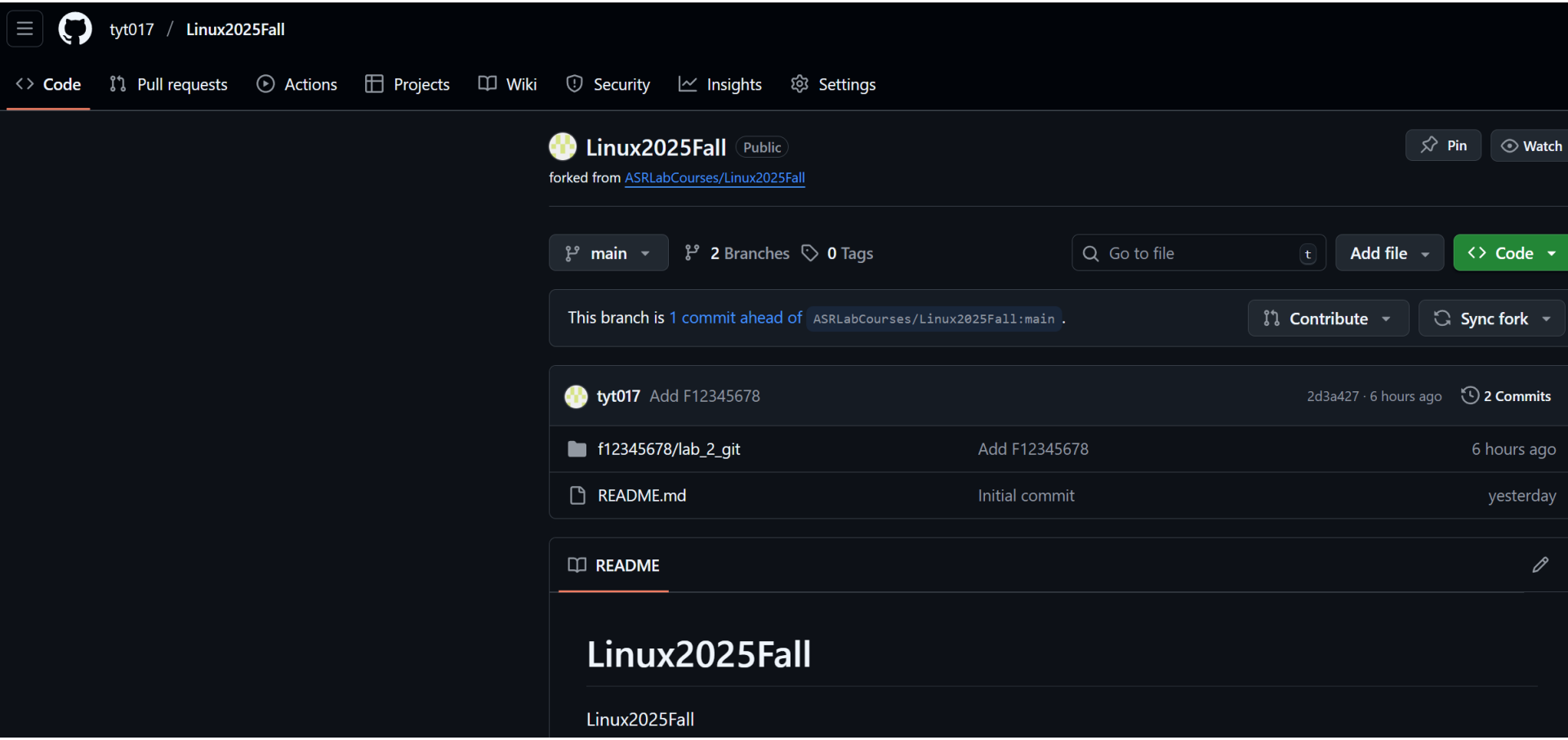
Merge your branch in local and push to remote

請參考上一頁的分支圖和老師上課投影片，依照下方圖中流程輸入對應的 git 指令來將 lab-2-git branch merge 到 main branch 並 push

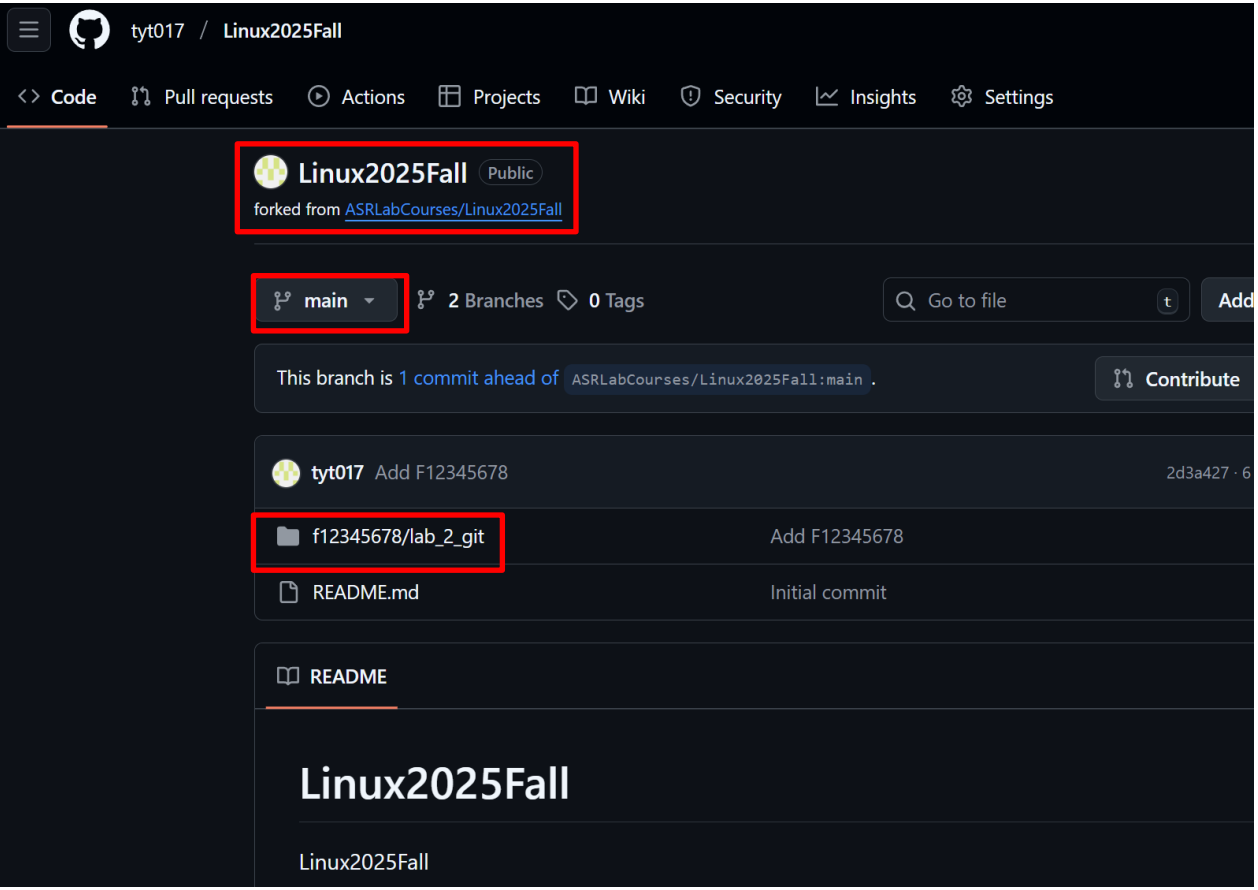
```
tyt@DESKTOP-EO1QCE5: /mnt
Total 5 (delta 0), reused 0 (delta 0), pack-reused 0
remote:
remote: Create a pull request for 'lab-2-git' on GitHub by visiting:
remote:      https://github.com/tyt017/Linux2025Fall/pull/new/lab-2-git
remote:
To https://github.com/tyt017/Linux2025Fall.git
 * [new branch]      lab-2-git -> lab-2-git
tyt@DESKTOP-EO1QCE5:/mnt/c/Users/USER/Desktop/Linux2025Fall/f12345678/lab_2_git$ cd ../..
tyt@DESKTOP-EO1QCE5:/mnt/c/Users/USER/Desktop/Linux2025Fall$ git [redacted]
M       README.md
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
tyt@DESKTOP-EO1QCE5:/mnt/c/Users/USER/Desktop/Linux2025Fall$ git [redacted]
  lab-2-git
* main
tyt@DESKTOP-EO1QCE5:/mnt/c/Users/USER/Desktop/Linux2025Fall$ git merge [redacted]
Updating 5ade602..2d3a427
Fast-forward
 f12345678/lab_2_git/first.txt | 1 +
 1 file changed, 1 insertion(+)
 create mode 100644 f12345678/lab_2_git/first.txt
tyt@DESKTOP-EO1QCE5:/mnt/c/Users/USER/Desktop/Linux2025Fall$ git push
Username for 'https://github.com': tyt017
Password for 'https://tyt017@github.com':
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/tyt017/Linux2025Fall.git
   5ade602..2d3a427  main -> main
tyt@DESKTOP-EO1QCE5:/mnt/c/Users/USER/Desktop/Linux2025Fall$
```

Open

<Your username>/Linux2025Fall



Demo



僅供參考

截圖檢查項目共兩項

1. 為 fork 的 repo.
2. 為 main 分支且 repo. 內有檔案

繳交作業需包含以下部分：

1. 將 github repo. 畫面截圖
2. 本次作業/課程心得

將上述內容彙整成一份 PDF 檔案繳交至 moodle

格式：**HW2_*StudentID*.pdf**
e.g., **HW2_F74131234.pdf**

Ref.

- Reusing a merged branch, good practice?
- Why Delete Old Git Branches?
- How can I save username and password in Git?

- SSH key tutorial: https://ithelp.ithome.com.tw/articles/10205988
- Step by step: https://aben20807.github.io/posts/20190421-github-flow-2/

QUESTIONS